The Effect of the CAHOOTS Shutdown on Eugene Police Response Times

DSCI 410 Data Science in Action Final Project Report – by Harrison Hoback

Background

After CAHOOTS, a Eugene alternative to emergency response, shut down in April 2025, questions arose about the potential impact on the efficiency of crisis response in Eugene. This research project attempts to answer some of those questions. More specifically, by analyzing police department response times after the shutdown and using Springfield, a city similar to Eugene, as a control, we can attempt to determine whether the police department responded to emergency calls differently and whether there is a causal relationship.

Response times may vary substantially between calls of different priority levels as higher priority calls require more urgency. Because of this, looking at how response times changed among different call priority levels could provide a finer level of detail about the effects of the shutdown and where this effect was seen the most.

This leads to one main research question and one subquestion:

1. How did Eugene Police Department (EPD) response times change when CAHOOTS stopped service?
    a. How did this differ among different call priority levels?

Data

Data Description

To answer these questions, two sets of emergency call data were used, one from Eugene and one from Springfield, containing data from 2015 to 2025:

<u>Eugene CAD Data (2015-2025):</u>

- **Source**: Excel file provided by Dr. Rori Rohlfs from a Eugene public records request.
- **Observations**: Individual call for service recorded by the CAD system.
- **Variables**: Year, agency, service, incident ID number, call time, case ID number, call source, nature, close code, closed as, seconds to dispatch, seconds to arrival, seconds to close, priority level, and vehicle prime unit
- **Strengths**: Comprehensive record of calls, a high level of detail about exact times and call type, a large dataset (sample size)

- **Limitations**: Missing or null values, a possibility for inaccurate times recorded, may exclude confounding variables that affect call times, provided as an Excel file (wrong variable types and hard to convert), and each year is not compiled together (harder to work with)

<u>SPD Calls for Service Data (2015-2025):</u>

- **Source**: Excel file provided by Dr. Rori Rohlfs from a Springfield public records request.
- **Observations**: Individual call for service recorded by the SPD system.
- **Variables**: Incident ID number, initial call type, final call type, responding agency, primary responding unit, call creation time, first dispatched time, first arrival time, clear time, priority level, and call creation mechanism.
- **Strengths**: Comprehensive record of calls, a high level of detail about exact times and call type, a large dataset (sample size), and each year is compiled together (easy to work with)
- **Limitations**: Missing or null values, a possibility for inaccurate times recorded, no year variable provided, provided as an Excel file (wrong variable types and hard to convert), and may exclude confounding variables that affect call times.

Data Cleaning & Preparation

The provided data was messy, unfiltered, and unusable as raw data. To use the data, it had to be cleaned, filtered, and combined into one dataset in preparation for analysis. This was done through a long, step-by-step process:

1. Explore the raw data by studying the format, missing values, variable types, any errors in the data, and determining variables of interest
2. Import each year of Eugene CAD and SPD data into R, compiling them into two lists by city
    a. SPD variable names were cleaned to deal with spacing errors and string formatting
3. Combine (stack) each list into two datasets by city (one for Eugene and Springfield)
    a. The priority column had to be converted into numeric values before combining, due to some values being listed as ‘P’ instead of numbers
4. Rename columns of both datasets to match each other for easier use and later combining
5. Clean the combined Springfield dataset:
    a. Convert times to numbers and then to datetimes
    b. Calculate the number of minutes to arrive by finding the time difference between the arrival time and the dispatch time
6. Clean the combined Eugene dataset:
    a. Convert times to datetimes and seconds to arrive/dispatch to numeric values
    b. Create a year column based on the year of call creation time
    c. Convert priority levels to numeric values
    d. Calculate the number of minutes to arrive by subtracting the dispatch time from the arrival time, dividing by 60 to convert seconds to minutes
7. Create a list of variables of interest from both datasets
8. Combine the Eugene and Springfield data into one dataset based on the common variables of interest

9. Reformat variables to fix spacing and string format issues, and convert all observations containing “NULL” to actual null values
10. Remove all calls self-reported by the police, filtering for only calls responded to by “EPD” or “SPD,” and removing all calls responded to by CAHOOTS
11. Remove outliers and unrealistic response times by only keeping values between 30 seconds and 6 hours
12. Filter out all priority levels higher than 5, since those calls do not reflect or require real response times
13. Remove any null values for year, agency, call time, priority level, and minutes to arrive
14. Create a monthly variable for easier visualization and analysis later
15. Save and export data cleaning steps to a CSV file

Scripts and Programs

All code scripts and programs used for this section are provided in the ‘data cleaning’ folder and explained in the corresponding ‘README.md’ file in the provided GitHub dashboard:

- https://github.com/hhoback/cahoots-response-times/tree/main

Methods

This study uses the fully cleaned and combined data to create a Difference-in-Differences (DiD) model to estimate the effect of the CAHOOTS shutdown on police response times in Eugene. A DiD model estimates the effect of a policy change or treatment by comparing the changes in an outcome over time between a treatment group affected by an intervention and an unaffected control group. In this analysis, Eugene serves as the treatment group because it experienced the CAHOOTS shutdown, while Springfield serves as the control group because it did not. The outcome or response variable is average police response times.

All analytical steps were completed in the same R Markdown file with the following input and output files:

- Input: “cleaned_eug_spd.csv” – the cleaned and combined Eugene and Springfield call data, completed in the “Data” section
- Output: “analysis.Rmd” – a single R Markdown file containing all analytical steps from each section of the analysis

The individual output files for each major analytical step are listed under their respective sections.

Packages

This analysis was performed in R using the following packages:

- tidyverse
- stats

- ggplot2
- fixest
- broom

Data Preparation and Initial Analysis

The purpose of this stage was to define new variables and filter the data to create datasets suitable for analysis and to visualize initial results of response time trends and differences between cities.

Steps

Priority levels were consolidated into three categories to simplify analysis and interpretation:

- Emergency (priorities 1-2)
- Investigative (priorities 3-4)
- Service (priority 5)

These categories are based on descriptions of each priority level provided by the City of Eugene: https://www.eugene-or.gov/DocumentCenter/View/530/Police-Call-Priority-Definitions

A treatment variable was created to separate calls between the treatment group (Eugene) and the control group (Springfield).

Two separate datasets were created for analysis of response times:

1. An overall dataset containing average response times, grouped by month and agency
2. A priority-group dataset containing average response times, grouped by month, agency, and priority group

For both datasets:

- Monthly average response times were calculated for each agency
- A variable of monthly call counts was included to provide information about the number of calls contributing to each average response time estimate and to provide context to later visualizations

An initial line plot was created to visualize the differences in response times between agencies and overall trends and changes over time

Outputs

- “plot_overall.png” – a line plot visualizing the pre-shutdown trends and post-shutdown changes in response times between Eugene and Springfield

Parallel Trends

For a Difference-in-Differences study to establish a causal effect, the data must satisfy the parallel trends assumption: Before the treatment (CAHOOTS shutdown), the treatment and control groups (Eugene and Springfield) must follow the same trends in the response variable (average monthly response times).

Each step below was done on the overall dataset and the priority group dataset:

Steps

Parallel trends were evaluated both visually and statistically. To visually evaluate it, average response times were plotted over time, with one line per agency (treatment/control), filtered only for pre-treatment data.

The statistical evaluation was done by estimating a linear regression model, testing whether pre-treatment response time trends were significantly different between Eugene and Springfield. The interaction term between treatment (agency) and month, along with the p-value, was used to determine whether the slopes were significantly different.

The regression results were summarized, tidied, and organized into a table for interpretational clarity. These results were compared with the visualizations to determine if the data passed parallel trends.

Inputs

- Monthly average response time dataset
- Monthly average response time by priority group dataset

Outputs

- "results_pt.csv" – csv file containing a table of all parallel trends regression results for the full dataset (p-values, coefficient estimates, etc.)
- "pt_facet.png" – a faceted line plot visualizing the pre-shutdown trends for each priority group
- "results_pt_pg.csv" – csv file containing a table of all parallel trends regression results for each priority group (p-values, coefficient estimates, etc.)

Difference-in-Differences

After checking for parallel trends, the main analysis was run using a Difference-in-Differences model to estimate the effect of the CAHOOTS shutdown on police response times.

Each step below was done on the overall dataset and the priority group dataset:

Steps

A linear regression model was estimated, testing whether post-treatment response time changes were significantly different between Eugene and Springfield. The interaction term between treatment (agency) and post-shutdown period represents the estimated effect of the shutdown on response times and was used, along with the p-value, to determine whether the changes in response times were significantly different.

The model used was:

- Response Time = $\beta_0 + \beta_1$*Treatment$_i$ + β_2*Post_Shutdown + β_3*Treatment$_i$*Post_Shutdown
  - Treatment is an indicator variable equal to 1 if the call was in Eugene or 0 for Springfield
  - Post_Shutdown is an indicator variable equal to 1 if the call was after the shutdown, and 0 if it was before

The model also controlled for month and agency, due to possible temporal or cross-agency differences in response times.

The regression results were summarized, tidied, and organized into a table for interpretational clarity.

Average response times were plotted over time, with one line per agency (treatment/control), and a vertical reference line at the shutdown date to separate pre-treatment and post-treatment trends.

Because the number of calls per priority group differs by city and by year, and can affect average response times, an area chart was created, plotting the frequency of calls per priority group for each city over time, providing context on sample sizes and prevalence of different call priority levels.

Inputs

- Monthly average response time dataset
- Monthly average response time by priority group dataset

Outputs

- "results_did.csv" – csv file containing a table of all difference-in-differences regression results for the full dataset (p-values, coefficient estimates, etc.)
- "results_did_pg.csv" – csv file containing a table of all difference-in-differences regression results for each priority group (p-values, coefficient estimates, etc.)
- "response_times_12.png" – a line plot visualizing the pre- and post-shutdown trends for the emergency priority (1-2) group
- "response_times_34.png" – a line plot visualizing the pre- and post-shutdown trends for the investigative priority (3-4) group
- "response_times_5.png" – a line plot visualizing the pre- and post-shutdown trends for the service priority (5) group
- "freq_pg.png" – an area chart showing the frequency of each priority group in each city over time

Scripts and Programs

All code scripts, programs, and results used for this section are provided in the ‘analysis’ folder and explained in the corresponding ‘README.md’ file in the provided GitHub repository:

- https://github.com/hhoback/cahoots-response-times/tree/main

Results

Initial Analysis

Figure 1 shows the monthly average of police response times in Eugene and Springfield over time. The x-axis shows years, but each point represents a month of the year, and the y-axis represents the average response time of that month in minutes.

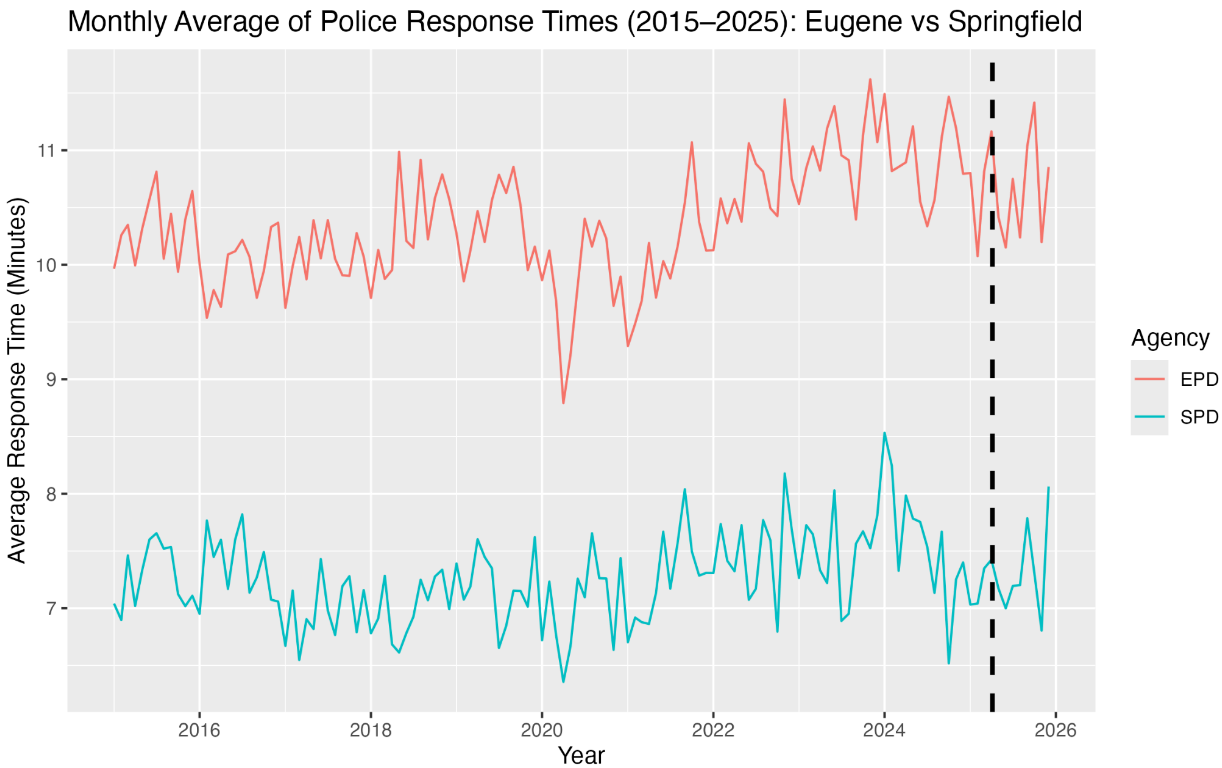

Figure 1

From this plot, we can see that Springfield has, on average, much slower police department response times. This plot was also used to visualize the parallel trends assumption for the difference-in-differences study of the full set of data.

Overall Response Times

Parallel Trends

Figure 2 is a table of regression results for the parallel trends statistical test of the full set of data. It contains two values of interest: the estimate of the interaction effect and the p-value, the estimate being -0.0001 and the p-value 0.003.

	term	estimate	std.error	statistic	p.value
1	(Intercept)	5.9232682697	6.212659e-01	9.5341911	1.614885e-18
2	year_month_num	0.0002427093	3.387541e-05	7.1647641	9.184968e-12
3	agencySPD	-0.4959991717	8.786027e-01	-0.5645318	5.729109e-01
4	year_month_num:agencySPD	-0.0001424761	4.790707e-05	-2.9740097	3.234083e-03

Figure 2

These regression results tell us that the magnitude of the pre-existing trends between Eugene and Springfield is close to 0, but this result is statistically significant since $p < 0.05$.

Difference-in-Differences

Figure 3a is a table of regression results for the difference-in-differences test of the full set of data. It contains two values of interest: the estimate of the interaction effect and the p-value, the estimate being 0.298 and the p-value 0.087.

	term	estimate	std.error	statistic	p.value
1	post_shutdown	-0.6491941	0.6760666	-0.9602516	0.33692902
2	treatmentTRUE:post_shutdown	0.2980527	0.1743161	1.7098407	0.08729591

Figure 3a

These regression results tell us that the estimated change in Eugene response times after the shutdown is around 0.3, meaning there is around an 18 second increase in response times, but since $p > 0.05$, the results are not statistically significant.

Response Times by Priority Group

Parallel Trends

Figure 3b shows the pre-shutdown trends of police response times in Eugene and Springfield, faceted by priority group, visualizing the parallel trends analysis of the data for each priority group. The x-axis shows years, but each point represents a month of the year, and the y-axis represents the average response time of that month in minutes.

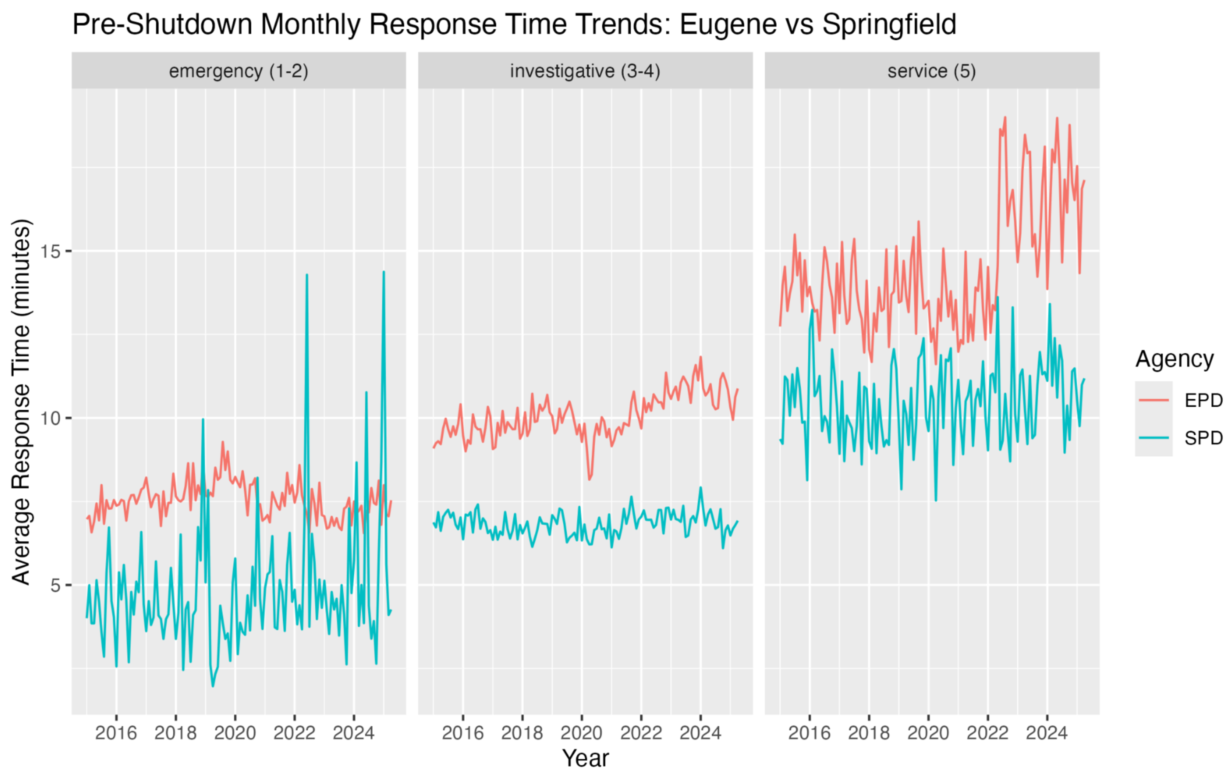

Figure 3b

Visually, the emergency group and the service group both clearly do not pass parallel trends. The investigative group is a little bit more subtle, but it looks like Eugene varies slightly more than Springfield, and doesn't pass parallel trends either.

Figure 3c is a table of regression results for the parallel trends statistical test for each priority group. It contains two values of interest per priority group: the estimate of the interaction effect and the p-value.

- For emergency calls (priorities 1-2): the estimate is 0.00046 and the p-value is 0.0049
- For investigative calls (priorities 3-4): the estimate is -0.00035 and the p-value is ~ 0.0
- For service calls (priority 5): the estimate is -0.00075 and the p-value is ~ 0.0

	priority_group	term	estimate	std.error	statistic	p.value
1	emergency (1–2)	(Intercept)	8.876153e+00	2.079935e+00	4.2675140	2.830590e−05
2	emergency (1–2)	year_month_num	−7.262349e−05	1.134114e−04	−0.6403542	5.225432e−01
3	emergency (1–2)	agencySPD	−1.109979e+01	2.941473e+00	−3.7735494	2.020998e−04
4	emergency (1–2)	year_month_num:agencySPD	4.558045e−04	1.603880e−04	2.8418863	4.864025e−03
5	investigative (3–4)	(Intercept)	2.934760e+00	6.529797e−01	4.4944124	1.078203e−05
6	investigative (3–4)	year_month_num	3.893999e−04	3.560465e−05	10.9367721	6.445473e−23
7	investigative (3–4)	agencySPD	3.242756e+00	9.234527e−01	3.5115559	5.305717e−04
8	investigative (3–4)	year_month_num:agencySPD	−3.537556e−04	5.035257e−05	−7.0255714	2.115340e−11
9	service (5)	(Intercept)	−2.448118e+00	2.064908e+00	−1.1855819	2.369408e−01
10	service (5)	year_month_num	9.245973e−04	1.125921e−04	8.2119228	1.272003e−14
11	service (5)	agencySPD	9.772252e+00	2.920221e+00	3.3464084	9.478599e−04
12	service (5)	year_month_num:agencySPD	−7.529053e−04	1.592292e−04	−4.7284368	3.828491e−06

Figure 3c

These regression results tell us that for every priority group, the magnitude of the pre-existing trends between Eugene and Springfield is close to 0, but each result is statistically significant since all p-values < 0.05.

Difference-in-Differences

Figure 4a is a table of regression results for the difference-in-differences test for each priority group. It contains two values of interest per priority group: the estimate of the interaction effect and the p-value.

- For emergency calls (priorities 1-2): the estimate is -0.76 and the p-value is 0.35
- For investigative calls (priorities 3-4): the estimate is 0.23 and the p-value is 0.21
- For service calls (priority 5): the estimate is 4.13 and the p-value is ~ 0.0

	priority_group	term	estimate	std.error	statistic	p.value
1	emergency (1–2)	post_shutdown	0.8589046	1.5164186	0.5664034	5.711230e–01
2	emergency (1–2)	treatmentTRUE:post_shutdown	–0.7633763	0.8112086	–0.9410358	3.466927e–01
3	investigative (3–4)	post_shutdown	–0.3935819	0.7193561	–0.5471308	5.842892e–01
4	investigative (3–4)	treatmentTRUE:post_shutdown	0.2305568	0.1833951	1.2571595	2.086966e–01
5	service (5)	post_shutdown	–4.0793914	2.6224137	–1.5555865	1.198110e–01
6	service (5)	treatmentTRUE:post_shutdown	4.1331716	0.7033648	5.8762841	4.214744e–09

Figure 4a

These regression results tell us that for emergency and investigative calls, the estimated changes in Eugene response times after the shutdown are around -0.76 and 0.23, meaning there is around an 46 second decrease and 14 second increase in response times, respectively, for each group, but since $p > 0.05$ for both, the results are not statistically significant. On the other hand, for service calls, the estimated change in Eugene response times after the shutdown is around 4.13, meaning there is around a 4 minute and 8 second increase in response times. Since $p < 0.05$, this result is statistically significant.

Figure 4b shows the monthly average of police response times in Eugene and Springfield for emergency (priority 1-2) calls over time. The x-axis shows years, but each point represents a month of the year, and the y-axis represents the average response time of that month in minutes.

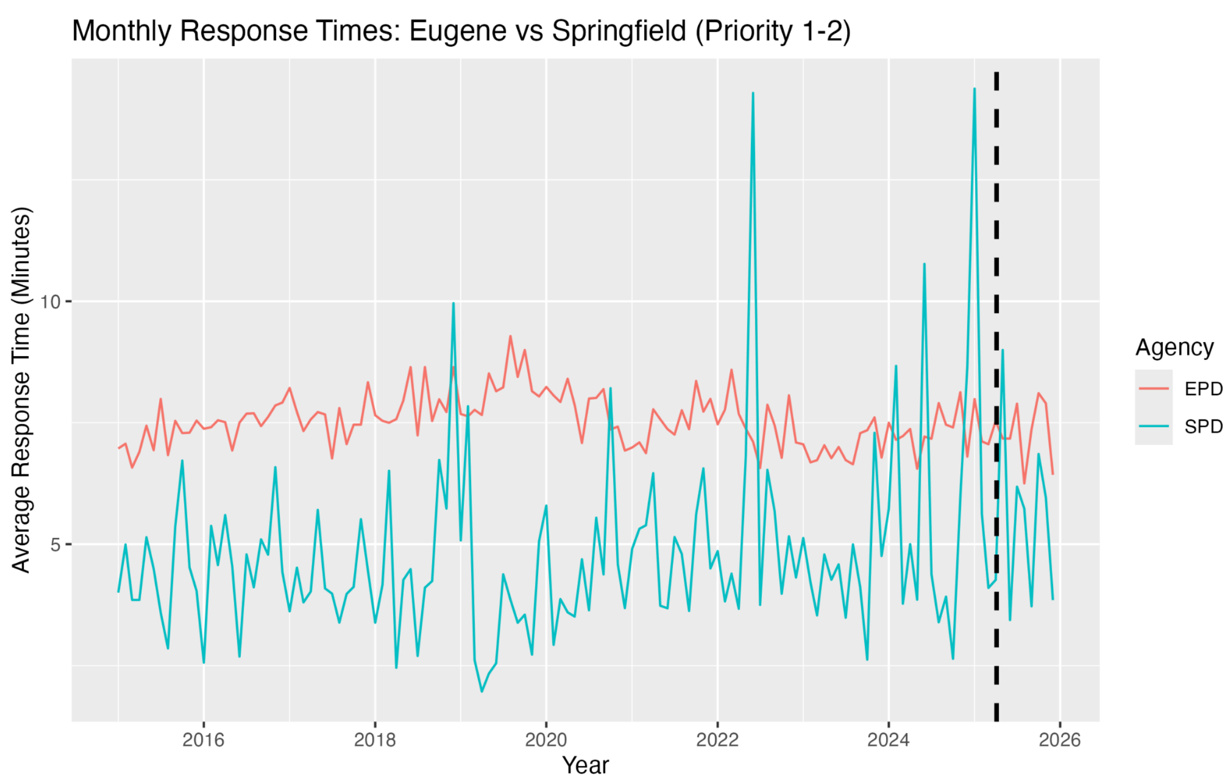

Figure 4b

Notably, there are many different spikes in the Springfield data. This is due to there being a very small number of emergency calls per month in Springfield after breaking the data down by month and priority group, so high values may skew the data.

Figure 4c shows the monthly average of police response times in Eugene and Springfield for investigative (priority 3-4) calls over time. The x-axis shows years, but each point represents a month of the year, and the y-axis represents the average response time of that month in minutes.

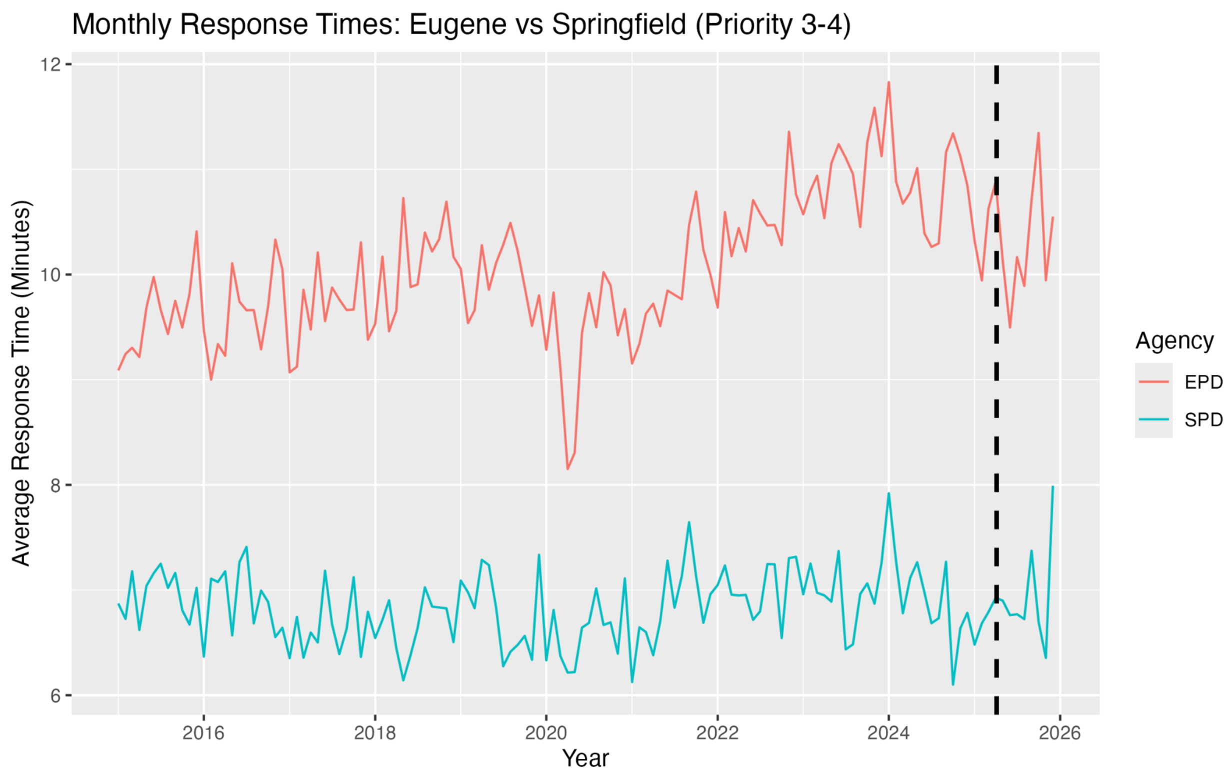

Figure 4c

We can see that this graph looks eerily similar to the original plot of overall response times. This is due to investigative calls being the large majority of call priorities that Springfield takes, at more than double than the amount of emergency and service calls combined, so the large sample size dominating the average response time calculation.

Figure 4d shows the monthly average of police response times in Eugene and Springfield for service (priority 5) calls over time. The x-axis shows years, but each point represents a month of the year, and the y-axis represents the average response time of that month in minutes.

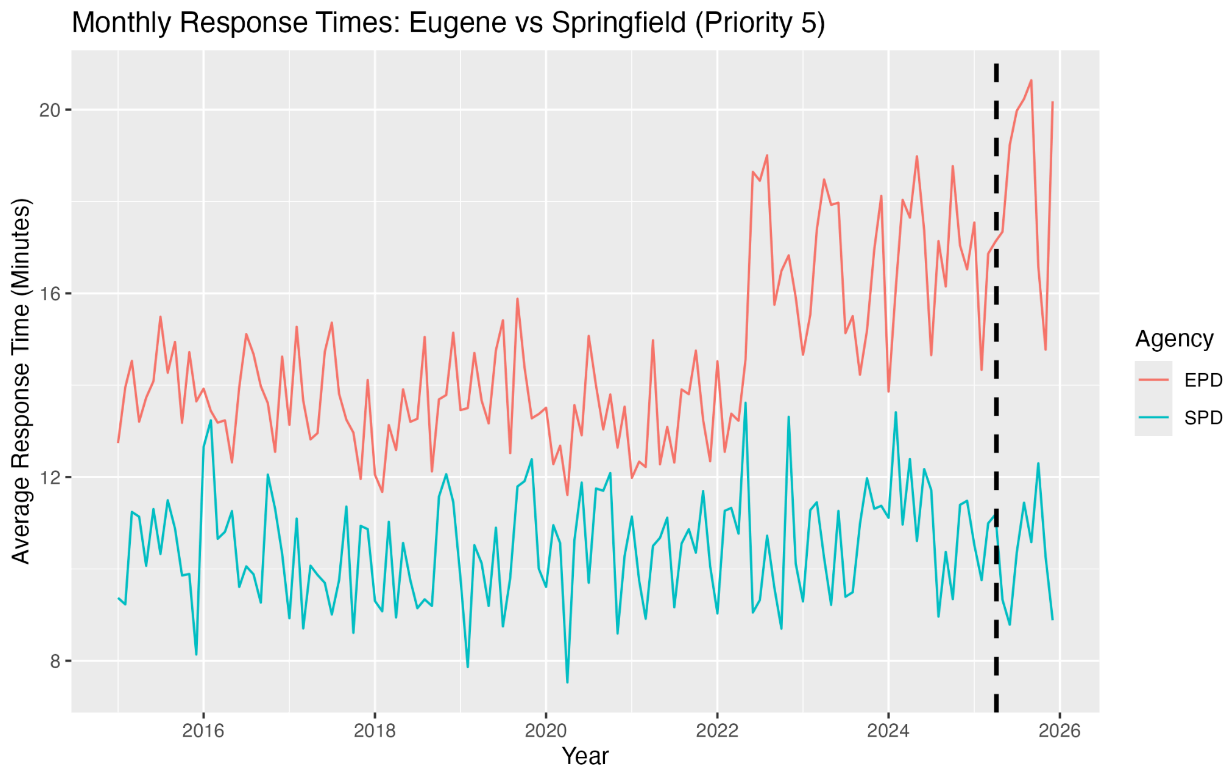

Figure 4d

A major overall increase can be seen in the Eugene response times after the shutdown, while Springfield stays roughly the same, if not a decrease. This matches the statistically significant increase of 4.13 we saw in Figure 3c.

Call Priority Frequency

Figure 5 shows the composition of call priority frequency over time, showing how many calls of each priority group are taken in each city and which calls are dominating the data, and which have a very small sample size.

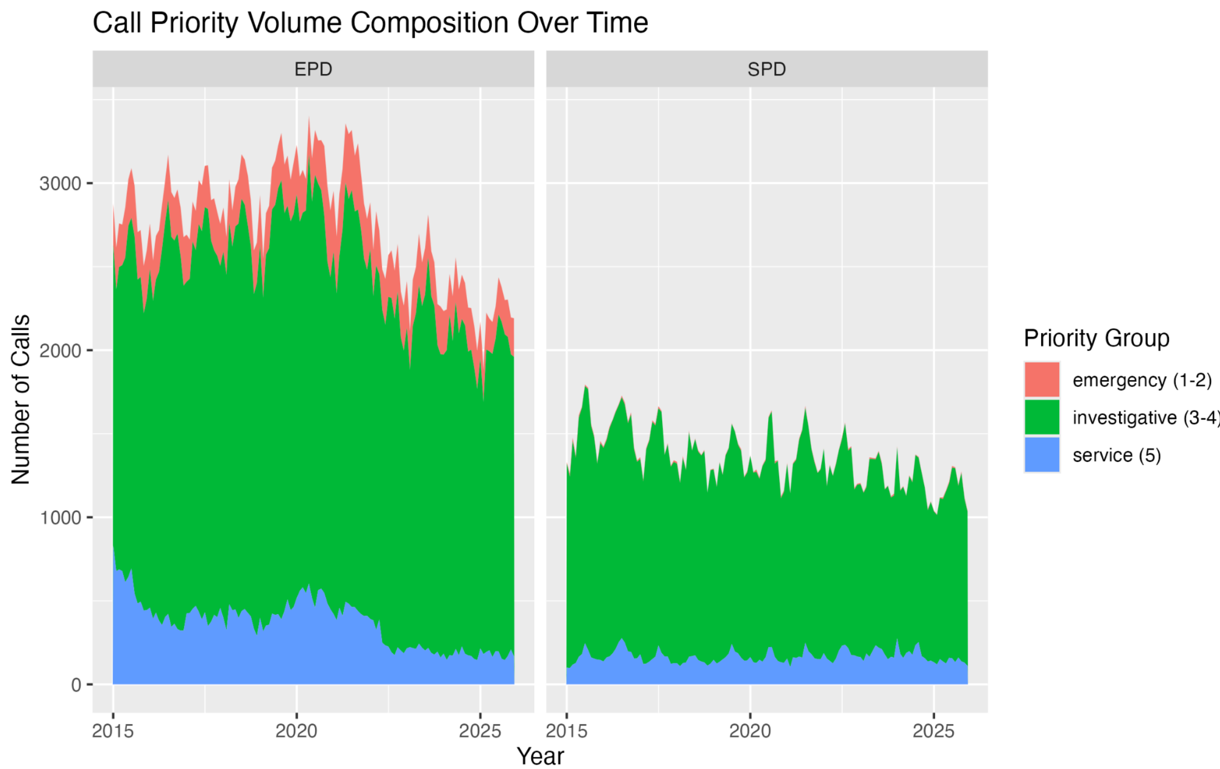

Figure 5

This figure provides context for the aforementioned caveats in figures 4b and 4c. The large majority of calls being taken in both cities, but especially Springfield, are investigative calls, so these ones are driving the average response times for each city. Emergency calls are seen the least and Eugene and rarely seen in Springfield, so the averages of this group can vary significantly and be skewed by high or low values.

Discussion

Key Findings

This analysis was conducted to determine whether or not there was a change in police response times in Eugene after the shutdown of CAHOOTS. After conducting this analysis, it was found that there was not a statistically significant change in response times after the shutdown when looking at all calls. After breaking it down by priority levels, however, some different results were found. Emergency calls (priority 1-2) and investigative calls (priority 3-4) were not found to have a statistically significant change in average response times after the shutdown, but service calls (priority 5) were found to have a statistically significant increase of around 4 minutes and 8 seconds. Because none of the priority groups passed the parallel trends assumption, however, this result cannot be interpreted as causal. While we can confidently say that there was an increase in response times for service calls after the shutdown, we don't know if this was caused by the shutdown.

Conceptually, these findings make a lot of sense. CAHOOTS was a community assistance program that primarily dealt with welfare checks and assisting individuals. They did not typically deal with life threatening emergencies or violent crime, which are priority 1-4 calls, so we would not expect to see a change in those response times. They did deal with welfare checks and service calls, which are priority 5, so the police likely had to deal with a higher proportion of these calls after the shutdown, slowing their response times to them.

These findings demonstrate the important of CAHOOTS as an alternative crisis response program to take the burden off the police. The increase in response times for service calls after the shutdown has a negative effect on the community, as they are receiving assistance slower and may have less trust in their institutions, or be unable to receive necessary care in a timely fashion.

Caveats

As mentioned in Figure 5, there is a very small number of emergency calls in the data relative to investigative calls and service calls. It's possible that this is resulting in inaccurate response times and resulting analysis. If this was taken into account when calculating average response times through weighting or another method, we might find different results.

There also seems to be a very significant and large increase in response times in early to mid-2022 that never seems to completely go back down, as all values afterward are on average, larger than previous values. The reason for this is unknown, and further research into how the Eugene Police Department records their response times or changes that were put in place around this time may inform this variation in the data.

Recommendations for Next Steps

This analysis was done two-fold: on the complete set of data with all priority levels, and broken down by priority group. For further analysis and to learn more about response times, doing this analysis on calls broken down by nature or call type could provide additional information about which specific types of calls are seeing changes in response times and what this might mean.

This analysis is limited by only having data for two cities. If we had more data about other Oregon cities or cities similar to Eugene that have CAHOOTS or a program similar, we could conduct a study using a synthetic control method, which is similar to Difference-in-Differences, but uses weights for multiple unaffected locations to better estimate the causal effect.

Additionally, this analysis is also limited by the amount of data we have. CAHOOTS shutdown a little over a year ago and we only have data for 8 months post-shutdown, while we have 10 years of pre-shutdown data. This ratio is disproportionate and the amount of post-shutdown data is very small to find meaningful changes that are statistically significant. This analysis could be redone multiple years from now with more data to find more significant and accurate results.